

Topics

N Puzzle Solvability

- To solve a puzzle, you are given an initial state and a goal state.
- Target is to reach goal state from initial state using puzzle rules.
- We can move blank tile only on left, right, up, and down.

6	1	10	2
7	11	4	14
5		9	15
8	12	13	3

Initial State

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

Goal State

- **Inversion** is a pair of tiles that are in the reverse order from where they ought to be. In our case, the bigger number is in front of the smaller one.

6	1	10	2
7	11	4	14
5		9	15
8	12	13	3

Initial State

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

Goal State

For tile-6, smaller tiles after it are-
tile-1, tile-2, tile-3, tile-4,
and tile-5, **total = 5**

Summation of inversion for
all tiles is inversion of the
puzzle.

$$\text{Inversion} = 5 + 0 + 7 + 0 + 3 + 5 + 1 + 6 + 1 + 2 + 4 + 1 + 1 + 1 + 0 = 37$$

Solvability Rules

- A puzzle is solvable if it satisfies one of the following three formulas:
 - If puzzle **dimension is odd** then puzzle is solvable only if the number of **inversion is even**.
 - If puzzle **dimension is even** and **blank tile is in odd row** from bottom then puzzle is solvable only if the number of **inversion is even**.
 - If puzzle **dimension is even** and **blank tile is in even row** from bottom then puzzle is solvable only if the number of **inversion is odd**.

Now, implement a code to check N Puzzle solvability of an initial board.

Topics

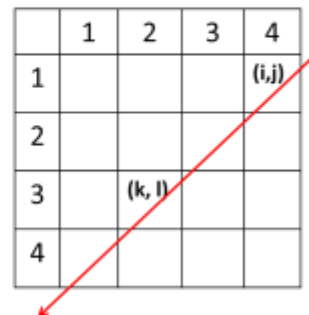
N Queen Attack Detection

Attack Types

- There are three types of attack:
 - Row attack
 - Column attack
 - Diagonal attack, two types of diagonal attack:
 - Upper left to lower right attack
 - Upper right to lower left attack

Detecting Diagonal Attack

- Left diagonal attack can be detected using -
$$i - j = k - l$$
$$\Rightarrow j - l = i - k$$
$$\Rightarrow j - l = -(k - i)$$
- Right diagonal attack can be detected using -
$$i + j = k + l$$
$$\Rightarrow j - l = k - i$$
- In general, we can write it as -
$$\text{abs}(j - l) = \text{abs}(k - i)$$



	1	2	3	4
1				(i, j)
2				
3		(k, l)		
4				

Now, implement a code detecting any type of attack in an arrangement.